

The Next Dimension

Emerging technologies that will change the way you see things on your display system:

3D TFT-LCD Monitor: Champagne served in a plastic cup somehow doesn't feel like Champagne at all. One could say the same for the kind of display you get when you combine the capacity of new-age graphics cards to render 3D objects in real time with plain Jane monitors.

3D TFT monitors will be here soon to complement today's advanced graphics cards. These monitors have a glass panel sandwiched between two LCD panels. Each LCD panel reflects half the light, cre-

ating a 3D sensation which takes realism to new heights.

OLED: Organic Light Emitting Diodes (OLED) are a new way to generate light, using organic materials rather than the complex crystalline structures found in traditional Light Emitting Diodes (LEDs). OLED technology produces self-luminous displays that do not require backlighting. As a result, you get thin, very compact displays.

OLED displays have many advantages over LCDs such as increased brightness, faster response time for full motion video, greater reliability and broader operating temperatures.

contrast, the performance of LG Flatron 782 LEs was quite mediocre—it could garner a score of just 8 points.

Viewsonic VP181 logged a top score of 16 in the 18-inch category. There was no visible colour tinge and no problem viewing the document at an angle. Surprisingly, Sony SDM-N80 showed a mediocre performance—the document could not even be seen clearly at an 80-degree angle and there was a noticeable yellow tinge.

Sharpness and Resolution

Here the test utility lights up only one pixel, which is made up of three subpixels representing red, green and blue. You can change the pixel as well as background colours at the press of a set of keys to see the impact of various colour combinations. If the pixels bleed into adjacent pixels, it indicates that the image generated will not be as perfect and sharp as it should be.

The 15-inch LCDs that did really well in this test were Philips 150 MT, Sharp LT-115V1 and Sharp LT-15S1. Philips 150 MT

scored over the two Sharp models with a hairline difference of just 0.67 points. There was no pixel bleeding in these three models and the display was crisp. An acceptable score here is 13 points.

In the 17-inch category, Samsung SyncMaster 171 MP and Viewsonic VE170 MB scored an impressive 16 points each. There was almost no noticeable bleeding in either model. By contrast, Easel CT-170 showed a lot of pixel bleeding and stood last with a score of just 4 points.

Sony SDM-N80 gave an outstanding performance in the 18-inch and above category, and scored 18 points, the highest across all categories. Viewsonic VP181 took a beating here, scoring only 8 points. The pixel dots were not perfect and a lot of pixel bleeding could be noticed.

Colour and Grayscale

LCD monitors reproduce colour more accurately and the brightness and contrast level is also far better than in CRT monitors. Another difference between LCD and CRT monitors is that LCD panels are made up of pixels which are individually lit.

Here, we had three subtests (midrange streaking, white level shift and black level shift). In real world applications, this would help a user judge how much clarity he can get in terms of brightness and contrast and whether there will be any ghost effect on the images.

Among the 15-inch monitors, Sharp LL-15V1 and Sharp LL-15S1 scored 11.16 and 10.83 points, respectively. A decent score here is 10 points. Neither monitor showed any streaking and ghosting effect, but the LL-15S1 showed a slight

Glossary

Active Matrix: An LCD technology used in flat panel computer displays. It produces a brighter and sharper display with a broader viewing angle than passive matrix screens. Active matrix technology uses a thin film transistor at each pixel and is often designated as a 'TFT' screen.

Aspect Ratio: The relationship of width and height. When an image is displayed on different screens, the aspect ratio must be kept the same to avoid stretching in either the vertical or horizontal direction. For most current monitors, this ratio is 4:3. For HDTV, the ratio is 16:9 or 16:10.

Colour Temperature: This defines the whiteness of the white colour on the screen. Natural colours used in life-like images, such as people or landscapes, look more true to life when displayed at a colour temperature of 6500K. Black text on a white page is better represented by a colour temperature of 9300K.

Dot Pitch: A measurement of distance between the centres of two same-colour phosphor dots on the screen. The closer the dots, the smaller the dot pitch, and sharper the image.

Ghosting: A visual effect in which an area of 'on' pixels causes a shadow on 'off' pixels in the same rows and columns. As a result, one can see two images of the viewable object.

Pixel: In a raster grid, the pixel is the smallest unit that can be addressed and given a colour or intensity. The pixel is represented by some number of bits (usually 8, 16 or 24) in the frame buffer, and is illuminated by a collection of phosphor dots in the CRT that are struck by the beams of the electron gun.

Enhanced Dot Pitch: This term refers to the Hitachi's tube technology in which the phosphor triads are spaced closer together horizontally than they are vertically.

white and black level shift. LG Flatron 568 LM was disappointing here too. A lot of ghosting and streaking was visible and its performance in the white and black level shift was also mediocre (it scored just 8.49 points).

In the 17-inch category, Viewsonic VE170 MB scored 11 points, just 1 point ahead of its nearest rival, LG Flatron 782 LE. Both showed minimal streaking and ghosting effect and their performance in the White and Black Level Shift test was also up to the mark. Another model that did well here was the Easel CT-170, with a score of 9.16 points.



The Viewsonic VP181 (18-inch LCD)